Convective Cloud Detection and Tracking Using the New-Generation Geostationary Satellite Over South China

Yikun Yang, Chuanfeng Zhao, Yue Sun, Yulei Chi, and Hao Fan

***Abstract*— As a core element of weather and climate change research, convective clouds have complex physical structures and dynamic processes, and knowledge of their evolution is limited by remote sensing data along with detection and tracking algorithms. In this study, we construct convective cloud detection and tracking algorithms for the Himawari-8 advanced Himawari-8 imager (AHI) data by combining machine learning models, area overlapping, and Kalman filter algorithms. First, we establish complicated and strict spatiotemporal data-matching conditions between the AHI and CloudSat Cloud Profile Radar (CPR) data, and a parallax correction model is developed to minimize the impact of parallax. To expand the sample volume of convective clouds, a region-growing algorithm is further used. Then, convective clouds during the day and night are detected based on the machine learning model which is trained through feature selection and hyper-parameter optimization. Finally, the automatic tracking of convective clouds, including those with relatively small scales or fast-moving speeds, is achieved by comprehensively using the area overlapping and Kalman filter algorithms. Validation results indicate that the algorithm developed can detect convective clouds of different scales with high accuracy, and the results show good continuity during day-to-night transitions. Moreover, the convective cloud detection algorithm can capture the convective clouds in advance compared with the traditional threshold algorithm. A case study of convective clouds demonstrates that the tracking algorithms we constructed can track convection at different scales. Overall, our study provides a new approach for convective cloud detection and tracking, which can contribute to a better understanding of weather and climate change.**

***Index Terms*— Convective cloud detection, convective cloud tracking, life cycle, machine learning.**

I. Introduction

CLOUDS are a crucial component of the Earth-atmosphere system, covering approximately 67% of the Earth's surface, and playing a critical role in the study of global weather and climate changes [1]. However, the simulations of clouds are challenging due to their large spatial and temporal variations and imperfect understanding of the mechanisms of occurrence and development. The current limited knowledge of aerosol-cloud interactions also hinders the understanding of cloud evolution. Additionally, various types of clouds exhibit significant differences in radiative and optical properties [2], and the net cloud radiative forcing at the top-of-atmosphere depends on both shortwave and longwave radiative forcing [3]. In general, low clouds typically have a negative net cloud radiative forcing, while high clouds can have either positive or negative net cloud radiative forcing.

Convective clouds are a special type of clouds that play a critical role in the global hydrological cycle and radiative energy budget at a local or global scale [4], [5]. These clouds are often organized into storm complexes, known as mesoscale convective systems (MCSs), accompanied by disastrous weather such as extreme rainfall and hail, which causes enormous casualties and economic losses [6], [7]. Previous studies have shown that MCS rainfall accounts for a substantial proportion of total precipitation [8], [9], [10]. However, the small spatial scale of convective clouds, along with their complex physical and dynamic processes and large spatiotemporal variability, contribute to significant uncertainties in existing weather and climate models' simulations of convective clouds [11]. In the past few decades, the scientific community has carried out many studies related to convective clouds, but most of them focus on the MCSs or deep convective clouds [4], [8], [12], [13].

To explore the properties of deep convective clouds, we first need to detect and track them. In terms of convective cloud detection, the most commonly used algorithms are based on fixed brightness temperature (BT) or radar reflectivity thresholds [8], [9], [12], [13]. Convective clouds, especially for deep convective clouds and MCSs, typically have a relatively low cloud top temperature (CTT), which makes them be detected by comprehensively using the BT and a suitable BT threshold with a central wavelength near 11 μm [14], [15], [16]. However, the BT threshold is not universal and spans a wide range, primarily due to the influence of atmospheric status, optical, macro-, or micro-physical properties of clouds, and spatial resolution of the data [17], [18]. BT difference (BTD) is also used for deep convective clouds or MCSs detection based on water vapor sensitivity and the difference between liquid and ice particle absorption spectra at different wavelengths [19], [20]. However, several factors affect the threshold of BTD, such as spatial and spectral resolutions, and signal-to-noise ratio [21], [22], [23]. Similar to the BT and BTD threshold algorithm based on satellite infrared channels, many threshold-based algorithms have been developed and

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used for convective cloud detection from ground-based radar data [9], [24], [25], [26]. Due to the dramatic changes in convective cloud characteristics from the initial to the dissipating stage, it is challenging for traditional threshold-based algorithms to achieve accurate detection in the whole life cycle. At present, most studies mainly focus on deep convective clouds, and there are large differences in the thresholds used for convective cloud detection [27], [28].

Without the need for understanding complex atmospheric physical and dynamic processes, the machine learning model has been gradually applied to atmospheric science in recent years to retrieve and estimate atmospheric parameters through data-driving approaches [24], [29], [30]. According to whether prior category knowledge is available for model adjustment and constraints during model training, convective cloud detection models can be divided into supervised classification and unsupervised classification. Unsupervised classification categorizes clouds into distinct clusters based on distance or similarity metrics [29], while supervised classification algorithms use more complex network structures to identify convective clouds [31], [32]. Using the cloud top pressure (CTP) and cloud optical depth (COD) dataset in the International Satellite Cloud Climatology Project (ISCCP) cloud products as characteristic parameters, Jakob and Tselioudis [33] and Rossow et al. [34] realized cloud type identification by K-means clustering algorithm over the tropical region. In the work of Berendes et al. [29], the standard deviation limited adaptive cluster (SDLAC) algorithm is proposed to identify the different types of cumulus clouds during the daytime, which adds a standard deviation threshold as a limiting condition and realizes the adjustment of the cluster centroid through iteration. The validation results show that it has less dependence on sensor spectral channels and spatial resolution. Although the K-means clustering and SDLAC algorithms can achieve cloud-type identification, they also have limitations, such as the need for subjective pre-definition of the number of cluster centroids and possible classification errors [35], [36], [37]. In recent years, supervised classification algorithms with high accuracy have been applied to identify convective clouds. For example, Lee et al. [38] achieved convective cloud detection in the daytime based on a convective cloud sample library established from the Multi-Radar Multi-Sensor (MRMS) precipitation type data. Although the MRMS data can be used to detect convective clouds, its ability to detect non-precipitating convective clouds is limited. The CloudSat Cloud Profile Radar (CPR) instrument can detect vertical structure of clouds, and its cloud classification products provide a reference for convective clouds detection. However, the construction of sample library for convective cloud detection using the CloudSat cloud type products is vulnerable to issues such as spatio-temporal matching errors and insufficient training samples.

The implementation of convective cloud tracking enables us to comprehend the changing characteristics throughout their life cycle and evolutionary mechanism. In recent years, numerous convective cloud tracking algorithms have been developed based on geostationary satellite or ground-based radar data. Among those algorithms, the area overlapping algorithm is relatively well-established [13], [39]. However, it is imperative to define the overlap rate of convective clouds at successive moments and the value of the overlap rate directly affects the accuracy of the tracking results. Additionally, the area overlapping algorithm is challenging to apply for the tracking of convective clouds that move at a fast speed or have a small spatial scale [18]. Even though some studies have improved the accuracy of the area overlapping algorithm to a certain extent by using the search radius as the judgment index [40], the threshold of the search radius also affects the tracking results due to the existence of certain differences in the movement speed of different convective clouds. The algorithm may fail when there is no overlap of convective clouds at adjacent moments or the overlap area does not meet the overlap rate threshold, particularly for coarse temporal resolution data, fast-moving convective clouds, or clouds with small spatial scales. Therefore, the area-overlapping algorithm is usually applied to the tracking of convective systems with larger spatial scales. Huang et al. [13], considering the Kalman filter algorithm can accurately predict the future state of the system when the initialization conditions contain errors, combined it with the area overlapping method to realize the mid-latitude MCS tracking, including small-scale or fast-moving MCSs. Chen et al. [12] improved the above algorithm and applied it to the Himawari-8 satellite observations to track the MCS in the Asian monsoon region, while the BT threshold algorithm is still used for MCSs detection.

A better understanding of cloud development and evolution is crucial in improving the simulation of weather and climate models related to convective clouds. The comprehension of the development and evolution of diurnal continuous multiscale convective clouds, which include cumulus and deep convection, is reliant upon their detection and tracking. Although the Himawari-8 geostationary satellite provides a database, there is a lack of corresponding convective cloud detection and tracking algorithms. Hence, this study aims to develop convective cloud detection and tracking algorithms based on the Himawari-8 geostationary satellite data, including both daytime and nighttime convective clouds. The remaining section of this article is organized as follows. The data used in this study is described in Section II. Section III provides an in-depth description of the convective cloud detection and tracking algorithms. The validation results of convective clouds are presented in Section IV. Section V summarizes the principal conclusions.

II. DATA

A. Himawari-8 Data

Himawari-8, a new-generation geostationary satellite, was launched on 7 October 2014, and operated by the Japan Meteorological Agency (JMA). The advanced Himawari-8 imager (AHI) instrument, onboard the Himawari-8 satellite, captures data from visible to thermal channels, including three visible, three near-infrared, and ten thermal channels [41], [42]. AHI data has a relatively high spatial resolution ranging from 0.5 to 2.0 km and a temporal resolution ranging from 2.5 to 10 mins compared with other operational geostationary satellite data [43]. Moreover, AHI data covers the East Asian monsoon region with longitudes ranging from 80° E to 160° W and latitudes ranging from 60° N to 60° S. Given the superior spatial resolution, temporal resolution, and spectrum channels, AHI data and its related retrieval products are widely used in Earth science research [44], [45], [46], [47], [48], [49].

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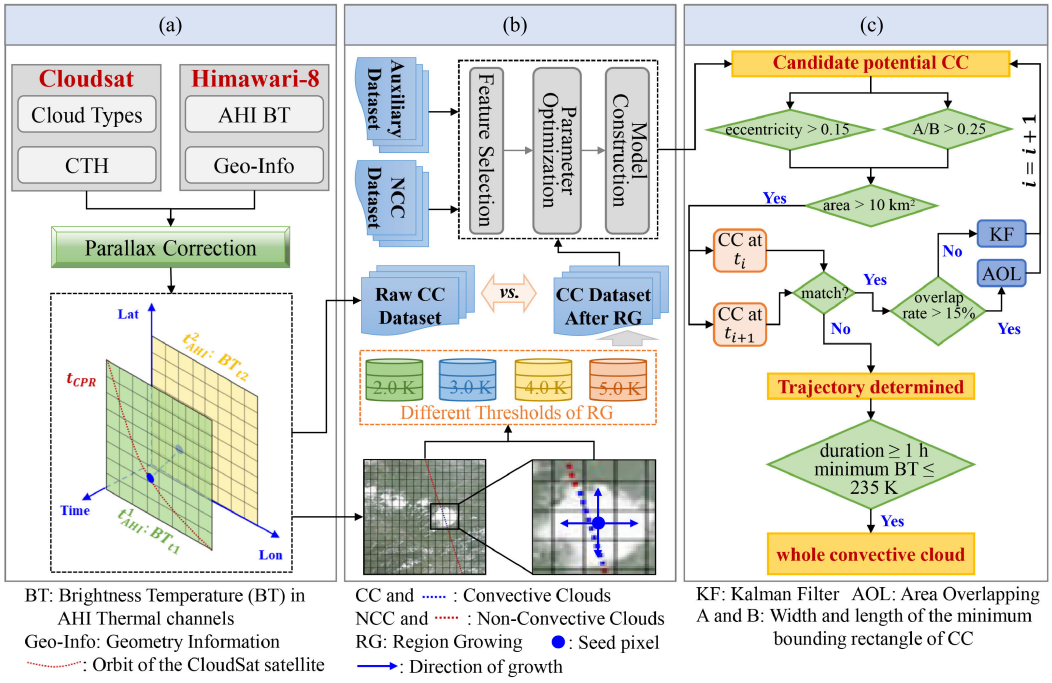


Fig. 1. Flowchart of convective cloud detection and tracking algorithms.

The high spatial and temporal resolutions of AHI data also satisfy the research requirements for convective cloud detection and tracking to a certain extent compared with other geostationary satellites. In this study, convective clouds mainly focused on east Asia with longitude ranging from 80° E to 135° E and latitudes ranging from 18° N to 55° N, which is consistent with the study region in our previous research. Due to larger time-consuming and calculation complexity, only a small fraction of convective clouds over the south China (104° E to 120° E and 22° N to 32° N) is tracked.

B. CloudSat Data

As an important satellite in the Earth Observation System (EOS), the CloudSat satellite is equipped with the first CPR with a frequency of 94 GHz [50]. The radar enables refined detection of the 3-D structure of clouds at a global scale [51]. In recent years, the data obtained by the CPR have been utilized for the statistical analysis of global cloud macro- and micro-physical parameters and the accuracy verification of cloud properties retrieved from passive satellite data or simulated by weather and climate models [52], [53], [54], [55].

To achieve convective cloud detection by combining the AHI data and machine learning algorithm, we collected the latest version (R05) of the level 2 cloud classification product from the CloudSat (hereinafter referred to as 2B-CLDCLASS). This product was developed using a rule-based algorithm that takes into account various cloud characteristics, including cloud height, temperature, maximum radar reflectivity factor, precipitation, as well as horizontal and vertical characteristics of clouds [56]. The 2B-CLDCLASS product categorizes clouds into cirrus, altostratus, altocumulus, stratus, stratocumulus, cumulus, nimbostratus, and deep convective clouds. Notably, only cumulus and deep convective clouds are marked as convective clouds, while the rest are marked as non-convective clouds.

C. Other Auxiliary Data

Terrain also has a certain influence on cloud formation, thus, the surface elevation data [i.e., digital elevation model (DEM)] as a feature is added to the convective cloud detection model. In addition, this study incorporates CTT and cloud top height (CTH) data, and determines the relative height of cloud (RCTH) between cloud top and surface based on DEM and CTH. It should be noted that CTT and CTH data are estimated based on the machine learning model. Compared with the CALIPSO CALIOP, MODIS, and JAXA AHI cloud products, the validation results of the CTT and CTH demonstrate higher accuracy. More details on the algorithm and validation results can be found in Yang et al. [44]. Additionally, near real-time seamless land surface temperature (LST) data from the ERA5 reanalysis data is collected. However, the spatial and temporal resolution of ERA5 reanalysis data is relatively low, so this study optimizes the spatial and temporal resolution of LST data using bilinear interpolation and inverse distance weighting algorithms, respectively.

III. Methodology

Fig. 1 illustrates the flowchart of convective cloud detection and tracking algorithms. First, multisource data matching, including cloud-type data, AHI BT data in ten thermal channels, geometry information, CTH, CTT, LST, and DEM, is conducted. Parallax correction is also implemented using the CTH obtained from the CloudSat data when matching with the Himawari-8 data. Second, the convective cloud dataset is expanded using a region-growing algorithm, and a machine learning-based convective cloud detection model is constructed. Finally, the shape factor and area of the convective cloud are calculated to determine the target convective cloud to be tracked. The area overlapping and Kalman filter algorithms are then combined to achieve the tracking of the convective cloud. A further detailed description of the methodology is described in Sections III-A–III-C.

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A. Data Matching

Traditional convective cloud detection algorithms are mainly based on BT thresholds, which can vary across different sensor channels and development stages, leading to greater uncertainty. To avoid the issue of threshold uncertainty in traditional algorithms, this study uses machine learning algorithms to detect convective clouds in combination with the CloudSat cloud types and AHI data. Several conditions are defined for data matching to reduce uncertainty in the training data of the machine learning model. First, the uppermost cloud type of the 2B-CLDCLASS data is selected as the final cloud type owing to the relatively weak penetration capability of the AHI sensor in clouds. Second, this study makes parallax correction based on the CTH obtained from the CPR cloud product to reduce the effect of parallax between the CPR and AHI sensors. Third, due to the rapid development or the relatively small spatial scale of convective clouds, especially the initial cumulus clouds, they may develop from a clear sky pixel to a convective cloud or move from convective clouds to a clear sky in a short time. For this reason, ± 1 min is selected as the time threshold in this study to limit the imaging time between the AHI and CPR data.

After the processes mentioned above, there are still some non-convective cloud samples that are marked as convective clouds, which will affect the detection results. Relative to the clear sky, convective clouds have a relatively low BT in thermal channels, and the BT of the cloud tops typically varies significantly with the development of convective clouds. Therefore, in the case of 2B-CLDCLASS data labeled as convective clouds, the multitemporal information of BT is further added to the matching of the CPR and AHI data, which improves the accuracy of data matching to some extent. The specific process is as follows: 1) extract the BT (BT_{t1}) of the AHI data at the moment (t_1) of matching with CPR and the BT (BT_{t2}) of the same position at the next moment (t_2) in the 10.4 μm channel and 2) when the absolute value of the BTD between the BT_{t1} and BT_{t2} is less than 0.5 K, the sample is considered more likely to be clear sky and the sample is removed.

However, when the sample is located at the edge of the convective cloud, the seed AHI pixel may be clear sky at the time t_1, but a convective cloud at the time t_2, and conditions (1) and (2) may not be able to effectively remove the false convective cloud sample. To solve the above problems, this study calculated the average value (μ) and standard deviation (δ) at the time t_1 of 3×3 windows around the seed pixel to describe the spatial uniformity. If the BT in the 10.4 μm channel of the seed sample is between $\mu - \delta$ and $\mu + \delta$, the seed sample is considered more likely to be a clear sky pixel at the time t_1. Based on the above rules, this study realized the matching of the AHI and 2B-CLDCLASS data from May to August 2016 at 80°–135° E and 18°–55° N. A total of 99, 567 valid samples are collected, of which convective cloud and non-convective cloud data account for 5.19% and 94.81%, respectively.

B. Convective Cloud Detection

Sufficient samples are a prerequisite for machine learning models with high accuracy and generalization ability. In this study, the convective cloud samples are expanded using the region-growing algorithm. Due to the obvious difference in BT between cumulus and deep convective clouds, the region-growing algorithm is executed separately for cumulus and deep convective clouds. Convective clouds are usually detected by using the BT threshold in the 10.4 or 11.2 μm channels [19], [57], [58]. Therefore, based on the BT in the 10.4 μm channel, using the convective cloud samples matched by the 2B-CLDCLASS and AHI as seeds, region growing is implemented by judging whether the BTD in the four-connected neighborhood satisfies the set threshold ($BTD_{10.4}$). Specifically, the BTs of the convective cloud seed and four connected neighborhood pixels are extracted, and the BTDs between four-connected neighborhood pixels and seed pixels are calculated. Then, the four-connected neighborhood pixels whose BTD is less than $BTD_{10.4}$ are added into the seed library, and the connected neighborhood pixel with the smallest BTD is selected as new seed. Finally, the region growing algorithm is terminated by iterating until the BTDs of four-connected neighborhoods of all seeds are greater than $BTD_{10.4}$.

The threshold $BTD_{10.4}$ is a key factor in the region-growing algorithm. In this study, four different thresholds (2.0, 3.0, 4.0, and 5.0 K) are set, and a reasonable threshold is determined by comparing the BT of convective cloud in each thermal channel before and after the region growing. Fig. 2 shows the BT distribution of cumulus clouds (left panel) and deep convective clouds (right panel) in the AHI thermal channels before (blue) and after (red) the region growing. For each box, the lower and upper lines represent the 25th (Q_1) and 75th (Q_3) percentiles of BT, and the height represents the interquartile range (IQR), where $\text{IQR} = Q_3 - Q_1$. The lower and upper whiskers are the $Q_1 - 1.5 \times \text{IQR}$ and $Q_3 + 1.5 \times \text{IQR}$, respectively. The middle line and point represent the median and mean values of BT.

Fig. 2 shows that the BT of deep convective clouds is significantly lower than that of cumulus clouds, and the BT distribution of convective clouds in different thermal channels is also different. Additionally, the BT distribution of the convective clouds before and after the region growing has a large difference with different thresholds set. A commonly used threshold for convective cloud detection is 235 K at the BT10 channel with a central wavelength of 10.4 μm. Most deep convective clouds can be extracted, while the detection capability of cumulus is limited. Generally, as the threshold increases from 2.0 to 4.0 K for cumulus clouds, the BT distribution (including the Q_1, Q_3, upper whisker, lower whisker, median, and average values) of most channels after the region growing matches better with the BT distribution of the original data. Even though the upper and lower whiskers of the result with the threshold of 5.0 K are more consistent with the original data compared with the results with a threshold of 4.0 K (e), the statistical indicators (such as average, median, Q_1, and Q_3) are slightly larger than that of the original data. Unlike the other channels, the BT07 after the region growing is higher than the original data, which may be affected by solar radiation. The solar zenith angle (SOZ) of convective cloud samples in the original data is greater than 30°, while the convective cloud dataset after the region growing includes many samples with SOZ less than 30°.

Different from cumulus clouds, the BT of deep convective clouds increases gradually with the increase of the threshold,

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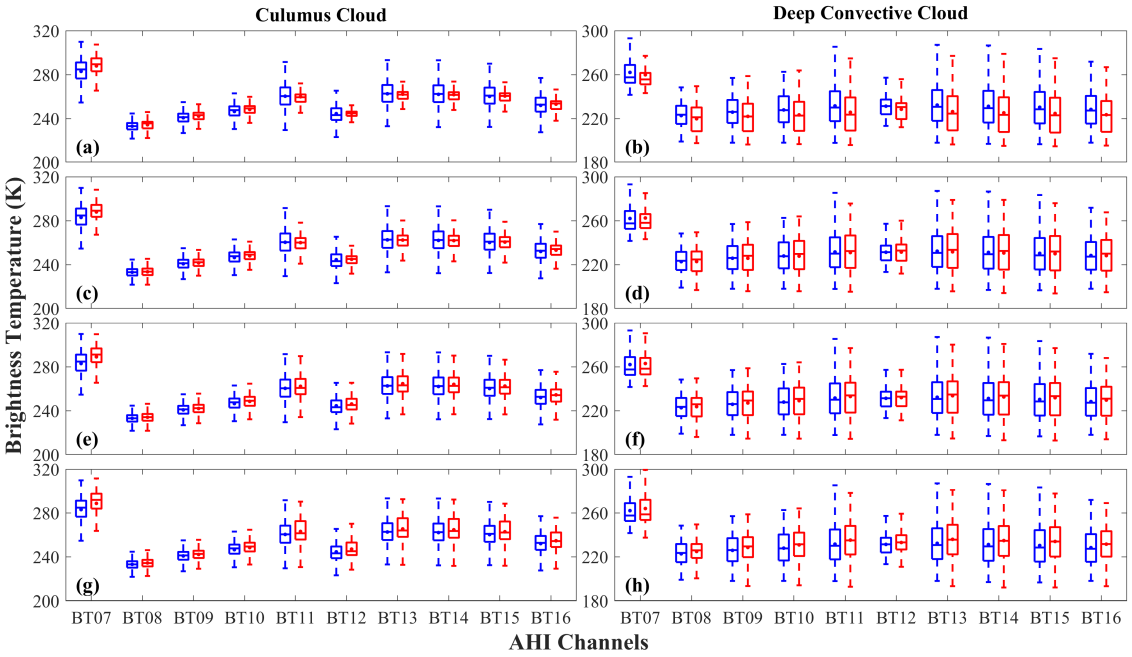


Fig. 2. BT boxplots of convective cloud in each AHI thermal channel before (blue) and after (red) the region growing. (a) and (b) 2.0, (c) and (d) 3.0, (e) and (f) 4.0, and (g) and (h) 5.0 K.

which is mainly due to the region-growing algorithm adding more samples with relatively higher BT to the dataset. When the threshold $BTD_{10.4}$ is 2.0 K, values of all statistical indicators (such as Q_1, Q_3, median, and mean values of BT) of BT obtained from the region growing algorithm are slightly lower than that of the original dataset. However, when the threshold $BTD_{10.4}$ is increased to 5.0 K, the values of indicators become greater than that of the original dataset. The BT distribution of the dataset after the region growing with 3.0 and 4.0 K as the threshold is similar to that of the original dataset. However, compared with the results with a threshold of 4.0 K, the IQRs of the results with a threshold of 3.0 K in each channel is slightly larger than that of the original data. Therefore, this study uses the convective cloud dataset obtained with a threshold of 4.0 K for model training, which makes the generalization ability of the convective cloud detection model stronger.

The XGBoost model is an ensemble machine learning algorithm widely used in the field of remote sensing [44], [59], [60]. Our previous research found that the XGBoost model has relatively high accuracy and efficacy in the retrieval of cloud macrophysical parameters compared with random forest (RF) and gradient-boosting decision tree (GBDT) models [44]. Therefore, we constructed a convective cloud detection algorithm using the XGBoost model. A total of 64 features, including thermal infrared BT, BTDs, CTH, CTT, LST, DEM, RCTH, BTDLST13 (the difference between the LST and BT13), and BTDLST14 (the difference between the LST and BT14), are selected as the initial features. To minimize the potential collinearity among features, the correlation matrix and feature importance score of all features are calculated first, and features with relatively low importance scores are removed referring to the iterative process described in our previous study [44]. Finally, 21 features are reserved for the construction of the convective cloud detection algorithm.

Hyper-parameter optimization is another key step that can improve the accuracy of the model and reduce the possibility of overfitting or underfitting. In this study, a total of 100 hyper-parameter combinations are tuned to obtain the optimal parameters for convective cloud detection. Specifically, n_estimator, max_depth, and learning rate range from 100 to 400 (in steps of 100), 4 to 20 (in steps of 4), and 0.01 to 0.17 (in steps of 0.04), respectively. Toward that end, the data is divided into the training dataset and validation dataset, where the training dataset includes the samples collected before the 25th of each month and validation samples collected from the 25th of each month to the end. The convective cloud detection model is constructed using the training dataset and used for the prediction in the validation dataset by continuously adjusting the parameter combinations. Then the accuracy of the model is analyzed by comparing the prediction label with the known label. Finally, the parameter combination with the highest accuracy in the validation dataset is selected as the optimal parameter and the importance score of each feature is re-calculated. Fig. 3(a) shows the importance score and cumulative importance score of the top seven features. BTD1516 has an important contribution to the detection of convective clouds with an importance score of 0.44, followed by BTD1415. The cumulative contribution of the top seven features can account for 86.0% of the total contribution. As shown in Fig. 3(b) and (c), the BTD values of channels 15 and 16 are relatively low, regardless of cumulus clouds and deep convective clouds, which may account for the relatively large importance score of BTD1516. Similar to convective clouds, cirrus clouds also have relatively low BTs; however, the AHI sensor lacks the cirrus channel (1.38 μm), which leads to great uncertainty in the removal of cirrus clouds. In this study, we refer to and improve previous research methods, and use BTD1415 greater than 2 K as the judgment condition to make further removal of cirrus clouds [61], [62].

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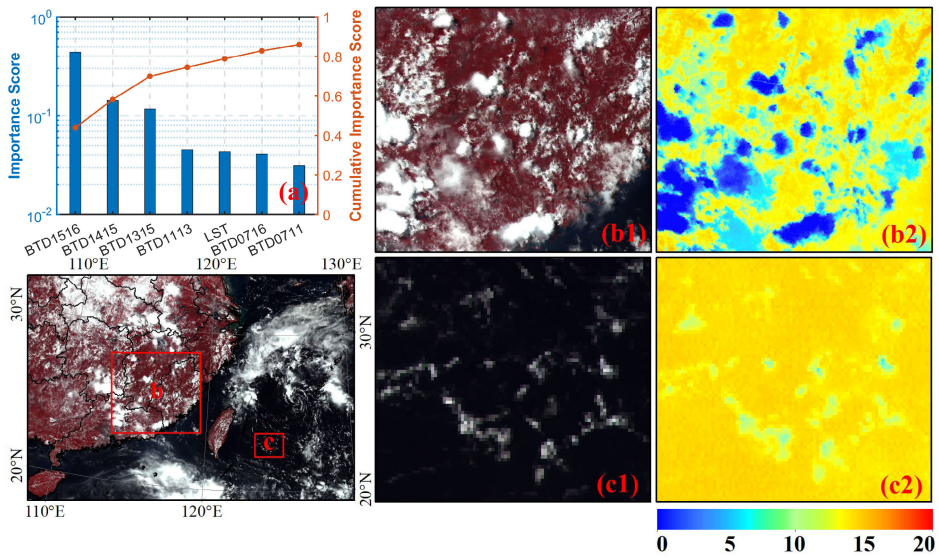


Fig. 3. Importance scores and the spatial distribution of BTD1516. (a) Importance scores and cumulative importance score of the top seven features. (b) and (c) Standard false-color composite image (RGB: 432) of convective cloud over land (b1) and ocean (c1) and the corresponding BTD1516 (b2 and c2).

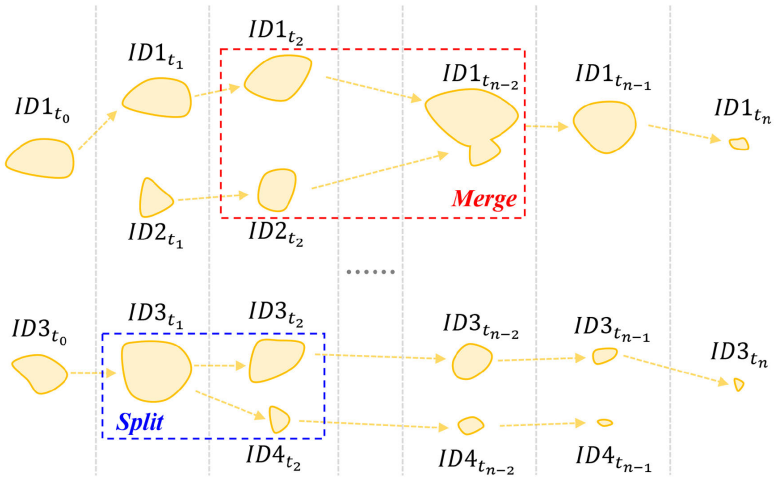


Fig. 4. Schematic illustration of the convective cloud tracking algorithm, each convective cloud is numbered in the form of IDX_{t_n}, and IDX_{t_n} represents the No. X individual convective cloud at the time t_n.

C. Convective Cloud Tracking

The area overlapping and Kalman filter algorithms have been used by previous studies to track MCSs from the Himawari-8 AHI geostationary satellite data [12], [13]. The Kalman filter mainly realizes the tracking of convective clouds with a relatively small scale or fast-moving speed, while the area overlapping algorithm mainly focuses on convective clouds tracking with larger areas. Different from previous studies, a machine learning algorithm is used in this study for convective cloud detection rather than a threshold-based algorithm, and some constraints are added to the tracking algorithm to improve the accuracy of convective cloud tracking.

As shown in Fig. 1(c), the convective cloud detection results are marked first as candidate potential convective clouds. Convective clouds usually have a circular or elliptical appearance, so this study further filters out non-convective by fitting the optimal ellipse and calculating the eccentricity. Since the cumulus clouds typically have a relatively small extent, it is hard for us to fit the optimal ellipse of cumulus clouds at certain times. Instead, the minimum enclosing rectangle is fit, and the ratio of width and length of the minimum enclosing rectangle is calculated. The detailed tracking method for convective clouds is summarized in Fig. 4 and shown as follows:

1) According to the detection results of convective cloud using a machine learning algorithm, the convective clouds at each moment in the study area are divided into different individual convective clouds based on the connectivity between pixels and their properties (including area, average BT, minimum BT, etc.) are calculated.
2) The eccentricity of the ellipse (defined as the ratio of the minor axis to the major axis of the ellipse) and the width-to-length ratio of the minimum enclosing rectangle are calculated if the ellipse or rectangle fits successfully. Only the convective clouds with an eccentricity greater than 0.15 or width-to-length ratio greater than 0.25 and area greater than 10 km^2 are tracked.
3) For convective cloud tracking, each individual convective cloud (such as IDX_{t_0}) at time t_0 is matched with

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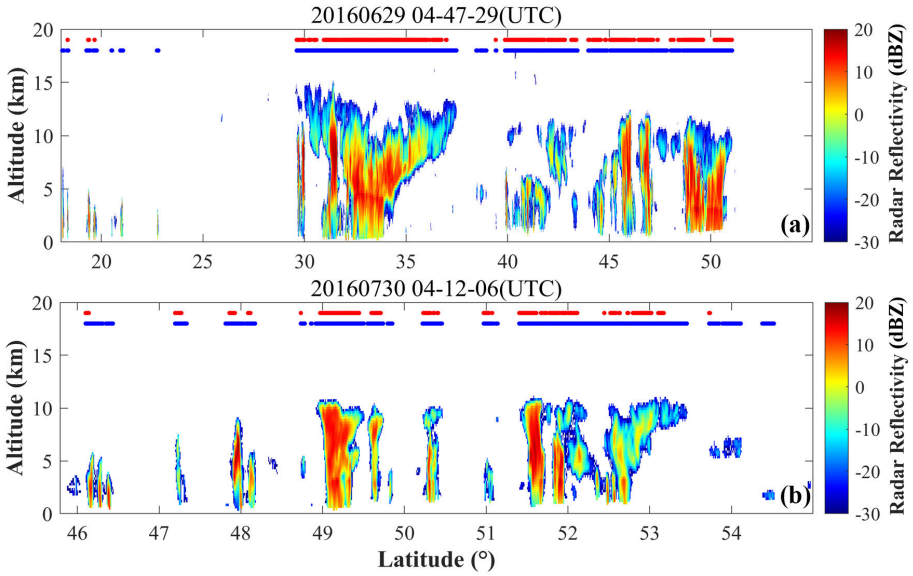


Fig. 5. Comparison of convective cloud detection result with CPR data. (a) June 29, 2016 and (b) July 30, 2016. The red and blue dots represent the echo of CPR and detected convective cloud, respectively.

the individual convective cloud (such as IDX_{t_1}) at time t_1. If IDX_{t_0} and IDX_{t_1} match successfully, the overlap rate is calculated first. If the overlap rate is less than 15%, the Kalman filter algorithm is selected for the convective cloud tracking, otherwise, the area overlapping algorithm is used. If there is no individual convective cloud matching with IDX_{t_0} at time t_1, the IDX_{t_0} is considered the last moment of the whole convective cloud life cycle.

4) The development of convective clouds is often accompanied by merging and/or splitting. In this study, the principle of maximum area is used as the criterion for judging the merging and splitting process of the convective clouds. As shown in Fig. 4, the red and blue boxes show the schematics of the convective cloud merging and splitting processes, respectively. $ID1_{t_{n-2}}$ is the merged result of $ID1_{t_2}$ and $ID2_{t_2}$. Since the area of $ID1_{t_2}$ is larger than that of $ID2_{t_2}$, $ID1_{t_2}$ is regarded as the predecessor of $ID1_{t_{n-2}}$. Similarly, $ID3_{t_2}$ inherits the properties of $ID3_{t_1}$ and continues to be tracked since the area of $ID4_{t_2}$ is smaller than that of $ID3_{t_2}$, and $ID4_{t_2}$ is tracked as a new individual convective cloud. Noted that the interval of t_{n-1} and t_n is 10 min in this study, which is consistent with the temporal resolution of the Himawari-8 geostationary meteorology satellite.

5) Constrained by the above rules, all convective clouds are tracked and the tracking trajectories are recorded. Two conditions are added to further filter the trajectory of convective clouds. First, there is at least one pixel overlap between adjacent convective clouds, and the other is that the area ratio of convective clouds at adjacent moment is greater than 0.2 and less than 5. In addition, to understand the properties of severe convective clouds, only the tracking trajectories with a duration greater than 1 h and the minimum BT in the whole life cycle less than 235 K are reserved.

IV. Results and Discussions

Based on the convective cloud detection algorithm that we constructed, the convective cloud is detected over East Asia. Fig. 5 represents the comparison result between the convective cloud detection result and the CloudSat CPR data on June 29, 2016 and July 30, 2016. As can be seen from the figure, the convective clouds detected by AHI data matched well with the CloudSat CPR data. The convective cloud detection algorithm constructed in this study can not only realize the detection of the larger-scale convective cloud system, but also has a certain ability to detect small-scale convective clouds. However, we should note that only the thermal infrared channel of the AHI data is used in the algorithm construction, so the existing convective cloud detection algorithm may misjudge large-scale cloud systems as convective clouds, especially in northern China. Overall, the convective cloud detection algorithm constructed based on a machine learning model can well capture the characteristics of deep convective clouds and also has a certain ability to capture shallow convective clouds. Furthermore, we also evaluate the accuracy of the model we have constructed in this study by using independent samples obtained from May to August in 2017. According to the statistical results, the model correctly detects convective and non-convective clouds with an accuracy of 59.6%.

Fig. 6 is an example of further verification of the detection results of convective clouds by machine learning (c) and (f) based on the cloud types from the JAXA AHI cloud product. It can be seen from the true color composite image (a) and (d) that deep convective clouds have large reflectance and spatial uniformity in visible channels, while shallow cumulus clouds have relatively small spatial scales and no obvious characteristics. In general, the convective cloud results from the JAXA AHI and machine learning have good spatial consistency with true color composite images based on visual interpretation.

With the development of convective clouds, the top of convective clouds reaches the bottom of the stratosphere, and the updraft is suppressed by the stratosphere and starts to spread

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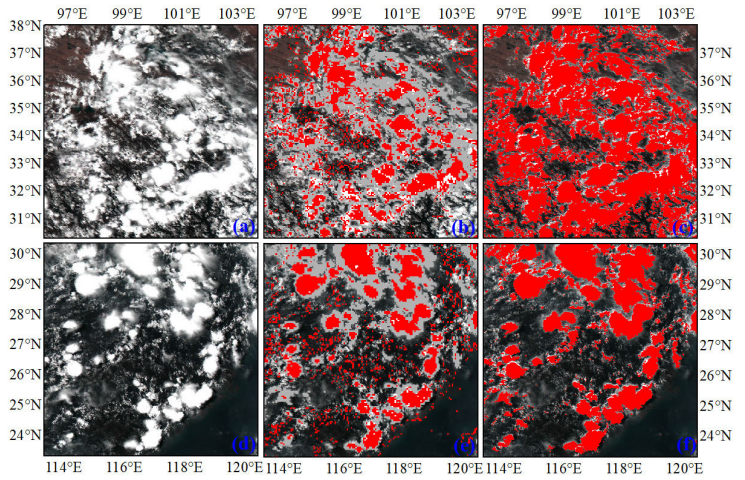


Fig. 6. Comparison of the JAXA AHI cloud types and convective cloud detection result. (a) and (d) True color composite image. (b) and (e) Cloud types obtained from the JAXA AHI cloud product. (c) and (f) Results of convective clouds detected by machine learning algorithm. Noted that the cirrostratus clouds (deep convective clouds and cumulus clouds) of the JAXA AHI cloud product are marked in gray (red).

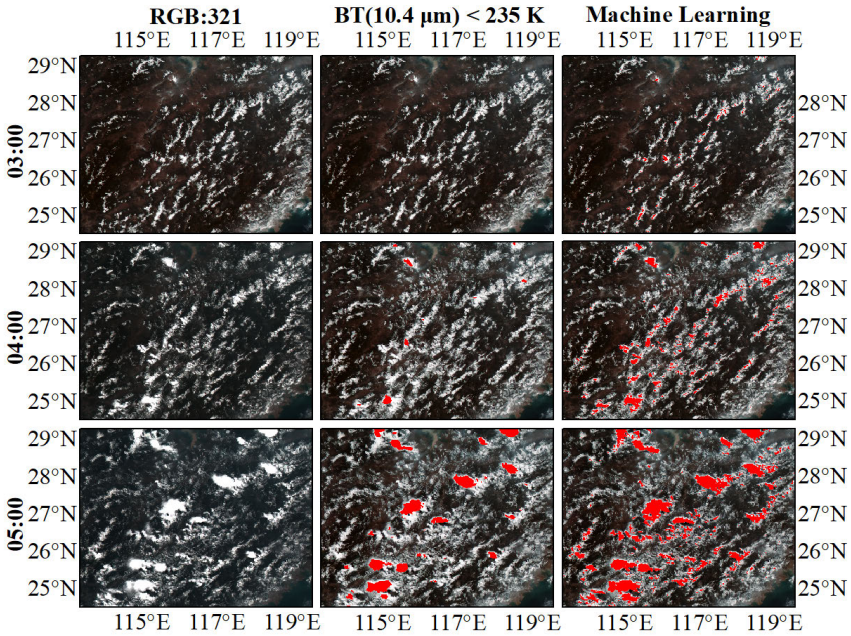


Fig. 7. Comparison of convective detection results between machine learning and traditional threshold algorithm. The convective clouds detected by the traditional threshold and machine learning algorithm are marked in red.

around, eventually forming a cloud anvil. Compared with the convective detection results derived by the machine learning algorithm, the JAXA AHI convective cloud mainly includes the core region of the convective cloud, while the algorithm constructed in this study marks the anvil also as a convective cloud. This is mainly due to the relatively low brightness of the anvil and the convective cloud core regions, which makes it difficult to distinguish the convective core and anvil when constructing a sample library based on the region growing algorithm. Additionally, the machine learning algorithm also has certain detection capabilities for shallow cumulus clouds. However, compared with the JAXA AHI results, the machine learning algorithm may miss some shallow cumulus clouds, which may be caused by the relatively small spatial scale of shallow cumulus clouds and the small number of samples in the convective cloud detection sample dataset. In addition, a convective cloud detection model based on the original samples (without region growing algorithm) is also constructed in this study, and the performance of two different models for convective cloud detection is compared. The results show that the convective cloud detection model constructed using the original samples can still mark some cloud anvils as convective clouds, and many small-scale convective clouds are lost.

The BT of 10.4 μm less than 235 K is usually used as the MCSs detection threshold in the traditional algorithms [12], [63]. Fig. 7 shows the convective cloud detection results in South China using different algorithms on July 30, 2016. Three rows represent three consecutive moments. It can be seen from the figure that as the SOZ decreases, convective clouds gradually develop from cumulus clouds

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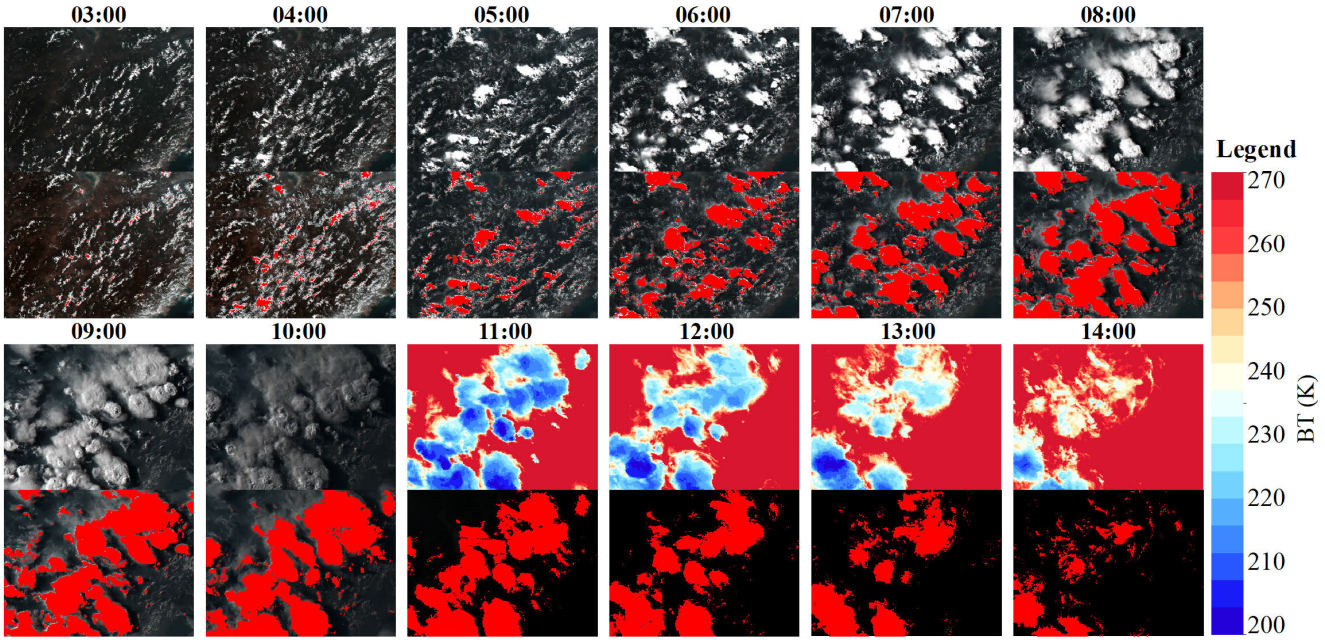


Fig. 8. Case analysis of convective cloud detection results in South China on July 30, 2016. In each subfigure, the upper panel represents the AHI true color composite image before 10:00 UTC and the BT of the 10.4 μm channel after 11:00 UTC, and the lower panel is the corresponding detected result of convective clouds.

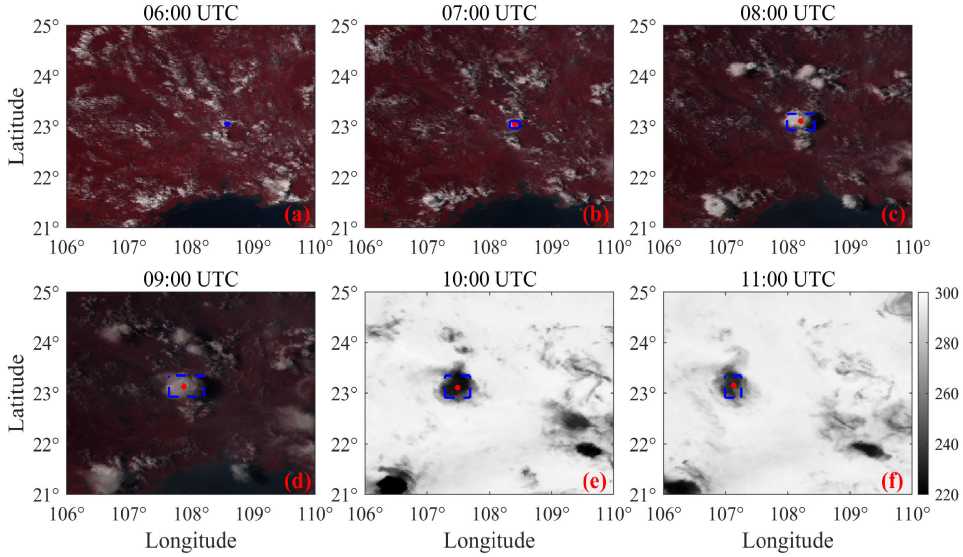


Fig. 9. Tracking case of convective cloud in South China on August 24, 2016, where the red dot and the blue box in the figure represent the geometric center and minimum enclosing rectangle of convective cloud, respectively. The background of subfigures (a)–(d) are filled with the AHI standard false-color composite image during the day, and the background image of subfigures (e)–(f) are BT with a central wavelength of 10.4 μm at night.

to deep convective clouds, and convective detection results derived from traditional threshold algorithms and machine learning have a high spatial consistency. However, the traditional threshold algorithm is greatly affected by the threshold value, causing many convective clouds in the initial stage to be missed, such as at 03:00 UTC (where the local time is 8 h ahead of the UTC time, similarly hereinafter), while the machine learning algorithm has a certain ability to detect small-scale convective clouds. Quantitatively, compared with the convective clouds detection results of traditional threshold algorithms, the convective clouds can be detected using the machine learning algorithm 1 h in advance, which allows us to get a more whole life cycle of convective clouds. A more whole life cycle of convective clouds is of great significance for the study of triggering prediction in weather or climate models.

Similar to the case of convective clouds in Fig. 7, Fig. 8 illustrates the evolution process of convective clouds in South China on July 30, 2016. It is evident that the development and evolution of convective clouds are closely related to surface downward shortwave radiation. With the accumulation of downward surface solar radiation, the development of convective clouds becomes increasingly vigorous. From 03:00 to 06:00 UTC, shallow convective clouds are triggered and developed with the aid of thermal or dynamical processes until they merge into a powerful convective complex.

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At 06:00 UTC, the cloud tops reach the top of the troposphere, and the convective clouds spread around to form anvil under the action of the strong updraft. The transition of convective clouds from the mature to the dissipating stage occurs at around 10:00 UTC.

Compared with true color composite or BT spatial distribution, the machine learning algorithm can better capture the whole evolution process of convective clouds, and convective detection results at night (after 10:00 UTC) have a high consistency with the spatial distribution of BT with a central wavelength of 10.4 μm. In addition, from the perspective of diurnal variation, the detection of convective clouds during the day and night can be obtained by using the BT of thermal infrared channels, and there is no significant difference between day and night. The results of convective cloud detection vary smoothly in the alternation of day and night (as in this case between 10:00 and 12:00 UTC), indicating that the convective cloud detection algorithm constructed in this study has high adaptability.

Based on the convective cloud detection and tracking algorithm we constructed, a total of 4, 359 convective cloud trajectories in the study region from June to August, 2016 are successfully recorded. Fig. 9 presents a convective cloud tracking case in South China on August 24, 2016. In this case, the convective cloud is triggered at around 06:00 (UTC), and disappeared at 11:40 (UTC), lasting for about 5 h and 40 min. The convective cloud detection and tracking algorithms constructed in this study can well capture the evolution processes of convective clouds. The area of convective clouds gradually expands from 40.96 km^2 (at 06:00 UTC) to 1828.62 km^2 (at 09:00 UTC), and the geometric center of convective cloud moves about 179.89 km from east to west during the whole life cycle, with an average moving speed of about 34.32 km/h. In addition, the cloud top BT also varies greatly during the evolution cycle of the convective cloud, with the average cloud top BT of 276.68 K at the initial stage, the minimum BT (lowest BT of all pixels with in whole life cycle) of 198.59 K (at 07:50 UTC), and the minimum average BT (lowest value of mean BT at all moments) of 220.35 K (at 08:30 UTC).

V. Summary and Conclusion

Convective clouds play a crucial role in the Earth-atmosphere energy budget and global hydrological cycle. Accurate convective cloud detection and tracking algorithms are the basis for studying the characteristics and evolution of convective clouds. In this study, we develop a convective cloud detection algorithm using a machine learning model based on the BT of the Himawari-8 AHI at thermal channels, CloudSat cloud-type data, and other auxiliary data. We minimize the uncertainties caused by the spatiotemporal match of multisource remote sensing satellite data by implementing a parallax correction model and strict data restrictions. Then, the region growing algorithm is used to expand the convective cloud dataset. Finally, a convective cloud detection algorithm based on the XGBoost model is trained with a series of processes, such as feature selection and hyper-parameter optimization.

Based on the convective cloud detection results, we further develop a convective cloud tracking algorithm that combines the area overlapping and Kalman filter algorithms. To minimize the influence of non-convective clouds, the shape factor of the convective clouds, including the eccentricity of the optimal ellipse and the ratio of width and length of the minimum enclosing rectangle, is calculated. Furthermore, according to the characteristics of the convective cloud, different tracking algorithms are used, ensuring the integrity cycle of the convective cloud. For the merging and splitting process of convective clouds, a tracking algorithm is introduced to determine the life cycle of the convective cloud by comparing the area of the convective clouds before and after merging or splitting.

The validation results of convective cloud detection indicate that the machine learning algorithm achieves accurate detection of convective clouds, including shallow cumulus clouds with relatively small spatial scales. The convective clouds obtained in this study have high spatial consistency with the CloudSat CPR and JAXA AHI cloud-type cloud products. Moreover, the detection algorithm can realize the detection of convective clouds during both day and night, and is less affected by SOZ. Compared with traditional threshold algorithms, convective cloud detection algorithms based on machine learning model can detect clouds at earlier stages. At the same time, by combining the area overlapping and Kalman filter algorithms, we can obtain a more whole convective cloud life cycle.

Although this study can well detect convective clouds with small-scale or fast-moving speed, due to the limited information in AHI thermal infrared channels, some clouds may be misjudged, such as anvil and larger-scale cloud systems. Object-oriented convective cloud detection algorithms that consider the spatial information of convective clouds may solve those problems.

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